GHraisoni INTERNATIONAL SKILL TECH UNIVERSITY Pune

G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination (CAE)

Programme: B Tech **Year: First** **Semester: I**
Course Code: MT101 **Max Marks: 20**
Course Name: Engineering Mathematics **Duration: 60 mins.**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting to answer.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	Apply Differentiation techniques for various mathematical functions
CO2	Apply Integration Techniques for various mathematical functions and find Area by using Integration

Q. No.	Sub Q. No.	Question	Marks
1		Solve any **TWO**	
	A	Find the equations for the tangent to the curve $y=5x^2-3x-6$ at the point $(1,-2)$. Use limit definition to find derivative	05
	B	Find c by Mean Value Theorem for the function $f(x) = 3x^2 - 5x + 1, g(x) = x$ defined in the interval $[2,5]$.	05
	C	It is given that the Rolle's Theorem holds good for the function $f(x) = x^3 + ax^2 + bx,\ x \in [1,2]$ at the point $x = \frac{4}{3}$. Find the values of a and b	05
	D	Find the critical points of $f(x) = (x^2 - 3)\,e^x$. Identify the open intervals on which f is increasing and decreasing. Find the function's local extreme values.	05
2		Solve any **TWO**	
	A	Evaluate the integral $I = \int cos^3(x) sinx dx$	05
	B	State the Fundamental Theorem of Calculus and apply it to find $\frac{d}{dx}(\int_0^x (3t^2 + 2t)dt)$	05
	C	Find the area under the curve $f(x) = x^2 - 4$ from $x = -2$ to $x = 2$	05
	D	Find the area under the curve $f(x) = cosx$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$	05